

A Low-Cost, Verifiable FPGA Resource for Embedded Machine-Learning Education in Open and Distance Learning

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Engineers must put models on real hardware — distance learners rarely can

- Modern curricula expect graduates to move a machine-learning model onto real silicon — training, numerical precision and digital logic together.
- This competence is built by doing, not by reading. You have to touch the hardware.
- In Open and Distance Learning (ODL), sustained laboratory access is the exception, not the rule.

For a distance learner, hands-on hardware is the exception, not the rule.

Embedded ML and hardware–software co-design are core skills — and they need real hardware.

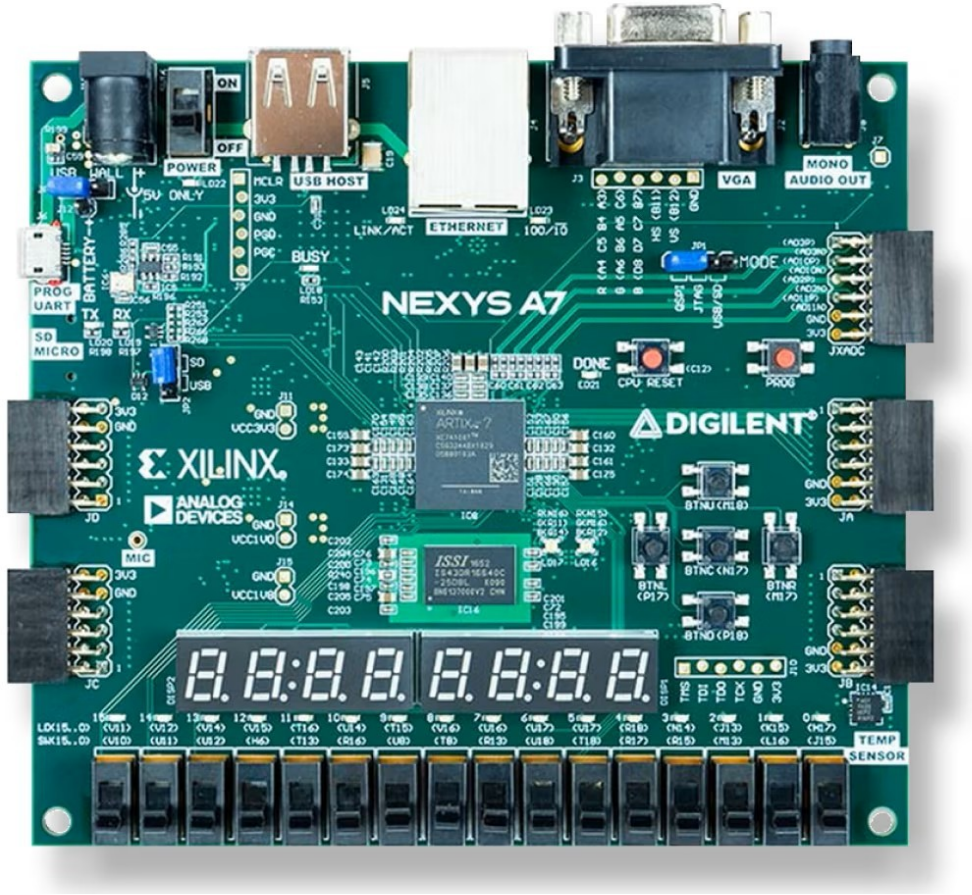
Why existing options fall short for distance learners

- Remote and virtual labs genuinely help — but they are institution-owned and time-shared, leaving little room for free, exploratory tinkering.
- Pure simulation is cheap — but it never lets the student confirm a design on real silicon.

Modality	Cost / learner	Owned by student	Real silicon	Built-in verification
Campus lab (hands-on)	High	No (shared)	Yes	—
Simulation only	Low	No	No	No
Remote laboratory	Medium (institutional)	No (shared)	Yes	Limited
This resource	**Low (≈ US$130)**	**Yes**	**Yes**	**Bit-exact**

Table I — No existing modality offers low cost, ownership, real silicon and built-in verification ***together.***

The proposed resource: a verifiable FPGA you own



Affordable & owned

- A commodity FPGA board (≈ US$130) with free vendor tools the learner owns it outright.

Full-stack in one project

- Training (machine learning) → fixed-point export (numerical methods) → RTL synthesis (digital design).

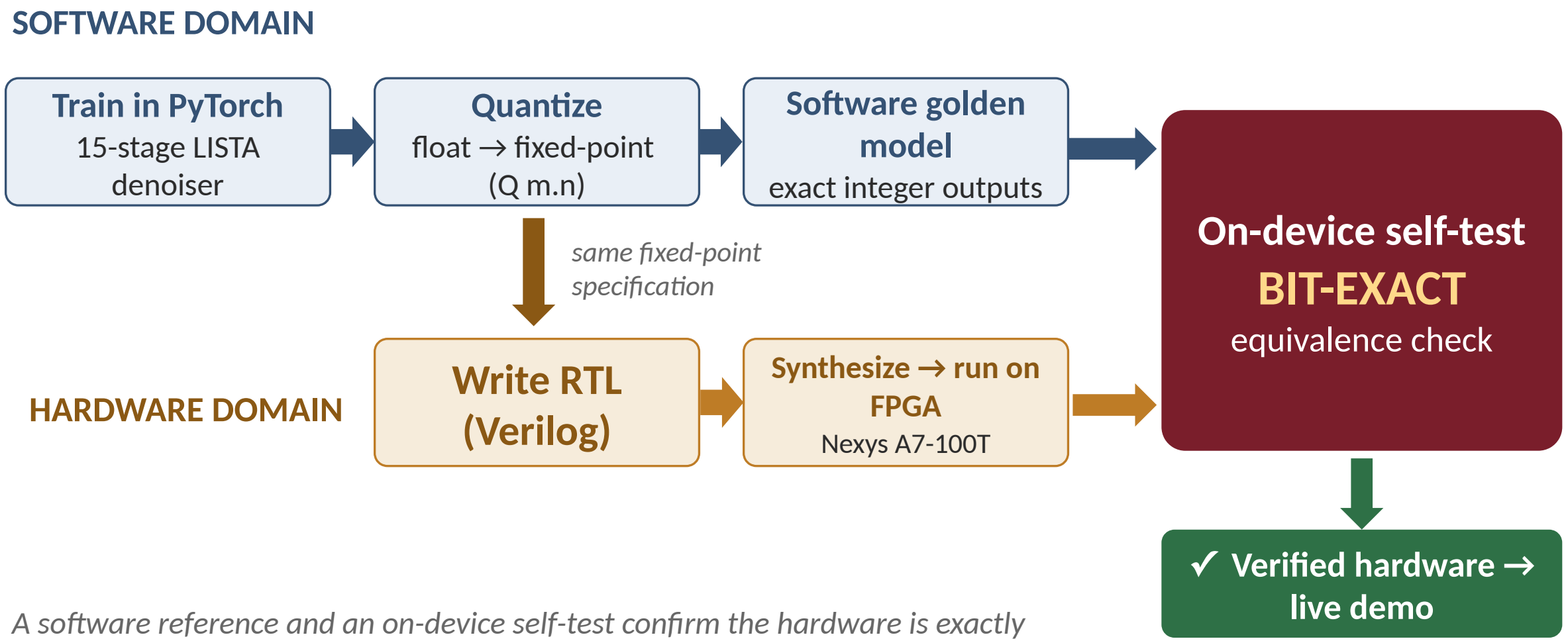
Verifiable by design

- An on-device self-test proves the hardware reproduces the software model, bit-for-bit.

Design-based research: an artifact built to be reproduced on a student's own desk.

Figure 1 — Nexys A7 AMD Xilinx Artix 7 FPGA (Mid range balanced performance FPGA)

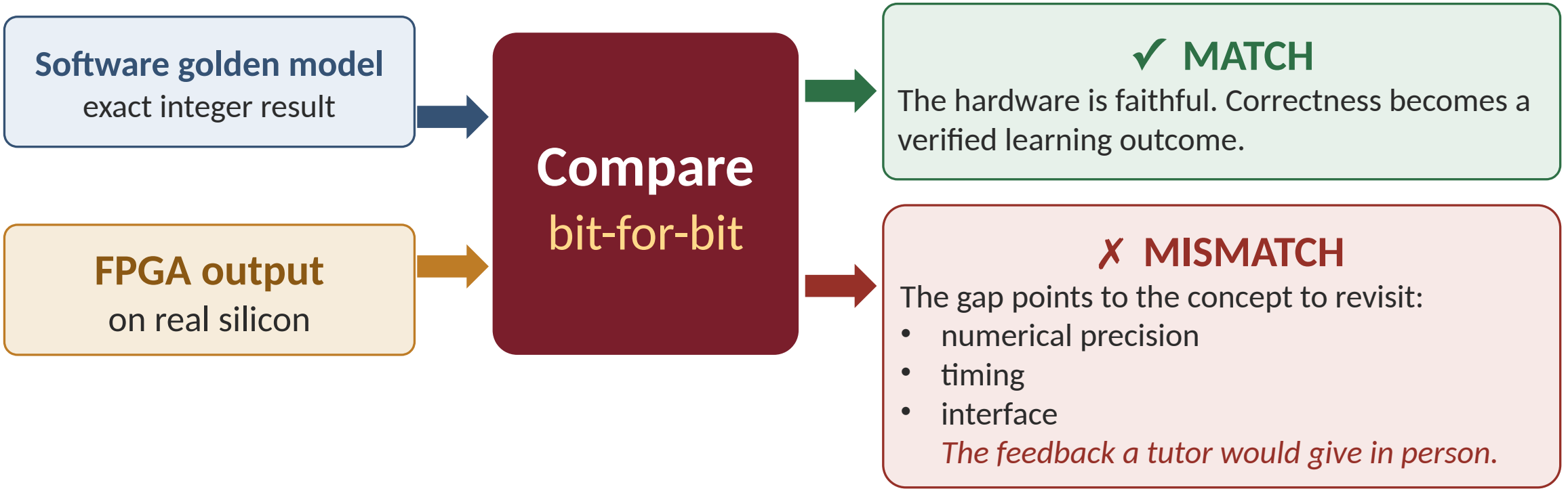
One pipeline: from trained model to verified hardware



A software reference and an on-device self-test confirm the hardware is exactly right — the same fixed-point specification drives both paths.

Bit-exact verification — the heart of the idea

A software “golden model” computes the exact integer result the hardware should produce.



Correctness becomes something the learner proves — not something a lecturer asserts. Debugging turns into guided learning.

Why embedded machine learning is the demonstration

- The hard, integrative skill — one task spans training, numerical precision and digital logic at once.
- Hard to teach remotely — it needs real hardware and real timing, not just a simulator.
- FPGA fits — it exposes the register-transfer-level mapping that microcontroller TinyML hides.
- The chosen example — a 15-stage LISTA sparse-coding denoiser: compact and interpretable in hardware.

One board, two domains, one identical datapath:

natural images + ECG signals

Denoising is a concrete task where 'better' is visible and measurable.

It works: one board, two domains

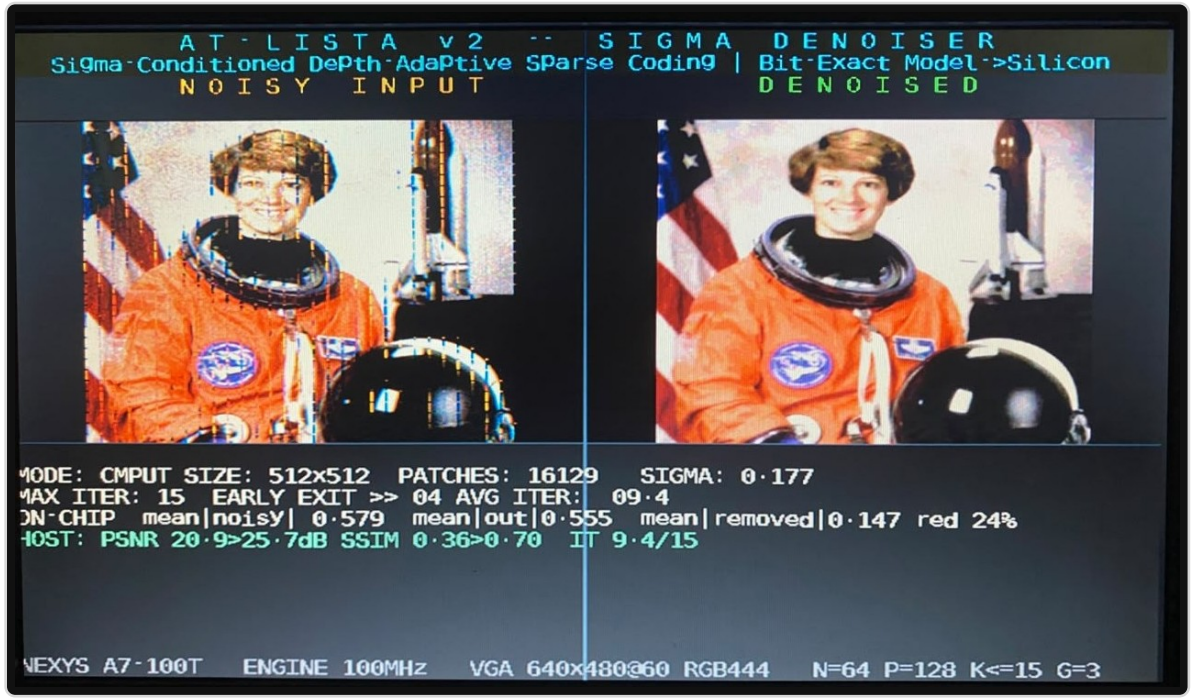


Figure 2-Natural image (512×512 RGB) — noisy → on-device denoised, live on VGA

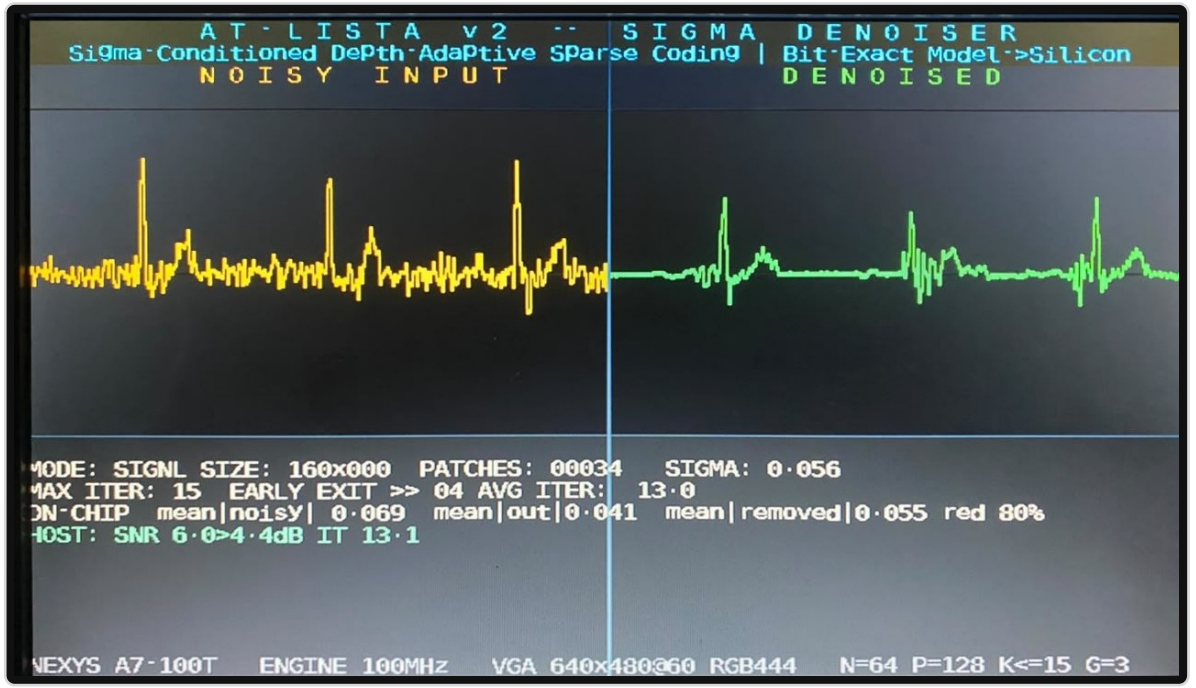


Figure3 - ECG signal — same board, same interface, live demo

Live on the Nexys A7-100T
100 MHz · VGA 640×480@60 · ≈63 patches/s

20.9 → 25.7 dB
image PSNR (+4.8 dB)

0.36 → 0.70
image SSIM

9.4 / 15
avg iters (early-exit)

Seeds, source and tooling are shared — a distance learner reproduces this and verifies it on their own board. *One illustrative run, not a benchmark.*

Verifiability as pedagogy — and what is new here

Staged learning (no tutor nearby)

- Each stage ships with a working reference; the self-test pins the rest.
- Rewrite one stage at a time — a failed check names what broke.
- Four-milestone rubric (in draft): reproduce the build → replace the golden model → diagnose a seeded precision fault → extend the network by one stage.

What is new — the contribution

- Educational: turning a correctness check into a learning outcome and assessment instrument.
- Engineering: a complete, open model-to-RTL pipeline on one low-cost board (images + ECG).
- Practical: reproducible, student-owned hardware that scales to a shared networked board.

The novelty is in the combination and the reframing of verification as pedagogy — not a new algorithm.

Limitations & what comes next

Limitations

- A feasibility / design study — no controlled student-outcome data yet.
- Still needs a host PC alongside the board.
- The example model is deliberately modest.

What comes next

- A controlled study across two cohorts — owned boards vs. one shared instance.
- A hosted remote-laboratory deployment over the network.
- A more capable model through the same verified flow.

The takeaway

This is not about building a better FPGA.

It is an educational resource that lets distance learners run authentic embedded machine-learning experiments — on affordable hardware they own, with correctness they can verify for themselves.

Affordable & owned · **Full-stack in one project** · **Bit-exact verified**

Thank you!

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